Supplementary Material

Deep Neural Network Surrogates for Approximate Bayesian Computation:
A Simulation Study for Mutation-Rate Estimation
in Markov Branching Processes

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This document contains the full architecture-, activation- and capacity-search tables for Study I (Section S1) and, for Study II, the full architecture search (S2), the GPS-ABC runs (S3), the check of our data generator against the training sets released by the original authors (S4) and the calibration rank histograms (S5). Each table below corresponds to a numbered claim in the main text; the surrounding prose there states the headline conclusion and is not repeated here except as needed to make each table self-contained.

S1 Study I: the full activation, normalisation and capacity search

The deployed one-dimensional surrogate (Section 4.1 of the main text) is a shallow funnel multilayer perceptron, input $\theta = \log_{10} p$, two GELU-activated dense layers at width 32–16 (626 parameters). That choice is the outcome of the two-stage search reported in full here: activation/normalisation first, at a 128–64 baseline, then capacity, once the activation question was settled.

Table S1: Architecture search, one-dimensional model: mean-curve fit MSE against the denoised response curve (irreducible replicate-noise floor $\approx 4.7 \times 10^{-3}$).

Model	Mean-curve MSE	vs. GP
Gaussian process (GPS-ABC baseline)	3.836×10^{-4}	—
DNN, GELU (128, 64), no BatchNorm [selects activation]	$\mathbf{3.888 \times 10^{-4}}$	+1%
DNN ensemble ×5, SiLU (128, 128, 64)	3.960×10^{-4}	+3%
DNN ensemble ×5, tanh (256, 128)	4.178×10^{-4}	+9%
DNN, SiLU (128, 128, 64)	4.400×10^{-4}	+15%
DNN, tanh (64, 64, 32)	4.435×10^{-4}	+16%
DNN, tanh (256, 128)	4.503×10^{-4}	+17%
DNN, ReLU + BatchNorm + dropout (64, 64, 32)	4.352×10^{-3}	+1035%

The row that determines every downstream comparison in the main text is the last one: an otherwise-reasonable network carrying BatchNorm and dropout fits *eleven times worse* than the same network without them. BatchNorm normalises per-minibatch statistics, useful in deep classification networks; but on a smooth scalar regression over a one-sided domain (the θ-grid has a hard edge at -8) it injects batch-dependent noise and biases predictions near that boundary. A follow-up ablation isolating only the activation function (same 128–64 architecture, no BatchNorm, averaged over three random seeds) found ReLU, SiLU, GELU, and tanh within 7% of one another

(3.95–4.23×10^{-4}): a statistical tie, with ReLU marginally ahead. Because accuracy does not separate them, the choice of GELU is made on structural grounds (smoothness, immunity to the dead-neuron failure mode) at no measurable accuracy cost. A second search round using 7-network deep ensembles (mean-curve MSE 3.94–4.25×10^{-4} across four ensemble configurations) did not improve on the single chosen network, so the simpler model is retained throughout.

Table S2: One-dimensional model: does capacity matter? Curve MSE is against the denoised response curve, mean over three seeds. Downstream MSE and 95% CI length are DNN-ABC at the same scale as the main text's mean-squared-error table (40 replicates/cell, full $p \times J$ grid), one seed per row.

Hidden layers	Parameters	Curve MSE ($\times$ GP)	Downstream MSE	Downstream 95% CI
GP (GPS-ABC baseline)	—	3.836×10^{-4} (—)	2.702×10^{-6}	2.137×10^{-3}
32–16 (deployed)	**626**	$\mathbf{3.896 \times 10^{-4}}$ (**+1.6%**)	$\mathbf{1.433 \times 10^{-6}}$	$\mathbf{1.300 \times 10^{-3}}$
128–64 (original)	8,642	4.129×10^{-4} (+7.6%)	1.514×10^{-6}	1.308×10^{-3}
64–32	2,274	3.957×10^{-4} (+3.2%)	1.565×10^{-6}	1.377×10^{-3}
16 (1 layer)	66	4.460×10^{-4} (+16.3%)	1.579×10^{-6}	1.375×10^{-3}
8 (1 layer)	34	4.983×10^{-4} (+29.9%)	1.668×10^{-6}	1.393×10^{-3}
16–8	186	4.508×10^{-4} (+17.5%)	1.810×10^{-6}	1.528×10^{-3}
Linear (control)	4	4.672×10^{-4} (+21.8%)	—	—

On raw curve fit, capacity is not merely unnecessary above a small threshold: it is mildly *counter-productive* at the original size: 128–64 has the largest gap to the GP of any two-layer candidate tested (+7.6%), while 32–16, at 14× fewer parameters, comes closest to matching it (+1.6%). On the downstream task the surrogate is actually used for, 32–16 is also the best point estimate on both accuracy and interval length of every candidate tested, 128–64 included, though attaching Monte Carlo standard errors to the gap shows this specific edge is not itself resolved (all nine Δ/SE values fall below 0.6 in magnitude). Nothing in the data argues against the smaller network either: 32–16 never does measurably worse than 128–64 in any cell. Only the linear control shows a clear, unambiguous cost, confirming some nonlinearity is required, just not much capacity beyond that.

S2 Study II: the full architecture search

Twelve architectures, five seeds each, on the four-parameter data (10,000 design points × 5 replicates; split by design point 80/10/10). Validation loss is the sum of the cross-entropy of the zero head and the Gaussian negative log-likelihood of the two features (standardised); the test columns are on held-out design points and were never used for selection. RMSE is in $\log_{10}$ units. Selection: best mean validation loss, then the cheapest model within one standard error of it.

S3 Study II: the GPS-ABC runs

GPS-ABC is the original authors' MATLAB implementation (`ABC_mu2a.m`, with their `mleHomGP` Gaussian-process code), unmodified, on our observed data sets, with the settings of Table S4. The only change is a memory-safe replacement of one line of the simulator, described in the main text (Study II, "The inference methods"): the original draws the second-stage cell count by summing `geornd` over a vector of about e^{τ} entries, which exhausts memory at the upper end of the τ box; a negative-binomial draw has the same distribution. The initial design is built once and shared by

Table S3: Study II: full architecture search (mean over five seeds; ± s.d. for the validation loss).

Surrogate	Params	Val. loss	Test NLL	RMSE $\log_{10} \bar{d}$	RMSE $\log_{10} s$	Zero log-loss	Train s	Epochs
ResNet 3×128 (GELU)	100,357	-3.3416 ± 0.0038	-2.901	0.1025	0.1076	0.0533	588	117
MLP 4×64 (GELU)	13,125	-3.3371 ± 0.0048	-2.907	0.1014	0.1074	0.0534	282	167
MLP 3×128 (SiLU)	34,309	-3.3280 ± 0.0032	-2.896	0.1015	0.1074	0.0539	343	160
MLP 256–128–64 (GELU)	42,757	-3.3211 ± 0.0041	-2.887	0.1021	0.1074	0.0536	213	115
ResNet 4×256 (GELU)	528,901	-3.2958 ± 0.0148	-2.856	0.1035	0.1079	0.0541	2342	96
MLP 128–64 (tanh)	9,221	-3.2924 ± 0.0104	-2.846	0.1047	0.1098	0.0553	296	207
MLP 128–64 (GELU)	9,221	-3.2895 ± 0.0131	-2.853	0.1073	0.1090	0.0545	285	193
MLP 64–32 (GELU/tanh)	2,565	-3.2841 ± 0.0112	-2.836	0.1049	0.1098	0.0554	261	251
MLP 128–64 (SiLU)	9,221	-3.2750 ± 0.0109	-2.836	0.1104	0.1106	0.0547	331	198
MLP 128–64 (ReLU)	9,221	-3.2702 ± 0.0156	-2.831	0.1043	0.1087	0.0544	228	150
MLP 32–16 (GELU)	773	-3.1927 ± 0.0318	-2.755	0.1138	0.1132	0.0569	233	271
MLP 16–16 (tanh)	437	-3.1499 ± 0.0301	-2.692	0.1127	0.1158	0.0596	223	295

all replicates, as in the authors' protocol. Chains are saved every 250 iterations by a copy of their sampler that differs only by that checkpoint (`patched/ABC_mu2a.m`), so a run can be analysed at the length it has reached.

Table S4: GPS-ABC settings (paper values; where the paper and the released code differ, the paper's).

Initial design	1,500 Latin-hypercube points × 5 single-culture draws
Refinement	10 new points when the unconditional error of the accept/reject decision exceeds $\xi = 0.3$
Surrogate draws per decision	$M = 100$; discrepancy $\epsilon = 10^{-8}$
Chain	target 20,000 draws, first 5,000 discarded; analysed here at 2,000 iterations, first 500 discar
Proposal step widths	$(0.8, 3.5, 3.5, 4)$ in the code's order $(\log_{10}\delta, \log_{10} p_1, \log_{10} p_2, \tau)$
Priors	$\mathrm{TN}(\log_{10} \hat{p}_{\mathrm{MOM}}, 20)$, $\mathrm{TN}(\log_{10} \hat{p}_{\mathrm{MOM}}, 20)$, $\mathrm{TN}(10, 20)$, $\mathrm{TN}(0, 20)$ on the paper box

S4 Study II: the data generator against the original authors' release

The test of the main text (Section 3.3), in full. At each design point of the training sets released with the original code, we draw 400 values from our simulator; the randomised probability-integral transform of each stored value is uniform if the generators agree. "Smallest bin p" is the smallest Kolmogorov–Smirnov p-value over quarters of each parameter's range (16 bins for the four-parameter set, 4 for each constant-rate set); at 1,500 draws per point the smallest four-parameter bin p-value rises from 0.005 to 0.08.

Table S5: Our generator against the training sets released by the original authors.

Released training set	Values	Zero frac.	Mean	Rep. sd	KS p (all)	Smallest bin p
4-parameter (init_2a.mat)	7,500	0.558	0.0075	0.0037	0.076	0.005
1-parameter p=1e-8 (init1_p8.mat)	15,000	0.386	0.0157	0.0053	0.220	0.182
1-parameter p=1e-6 (init1_p6.mat)	15,000	0.356	0.0508	0.0163	0.522	0.077
1-parameter p=1e-4 (init1_p4.mat)	15,000	0.178	0.1339	0.0551	0.692	0.264

S5 Study II: calibration rank histograms

Ranks of the true parameter among thinned posterior draws, over 500 truths drawn from the prior. A calibrated posterior gives a flat histogram.

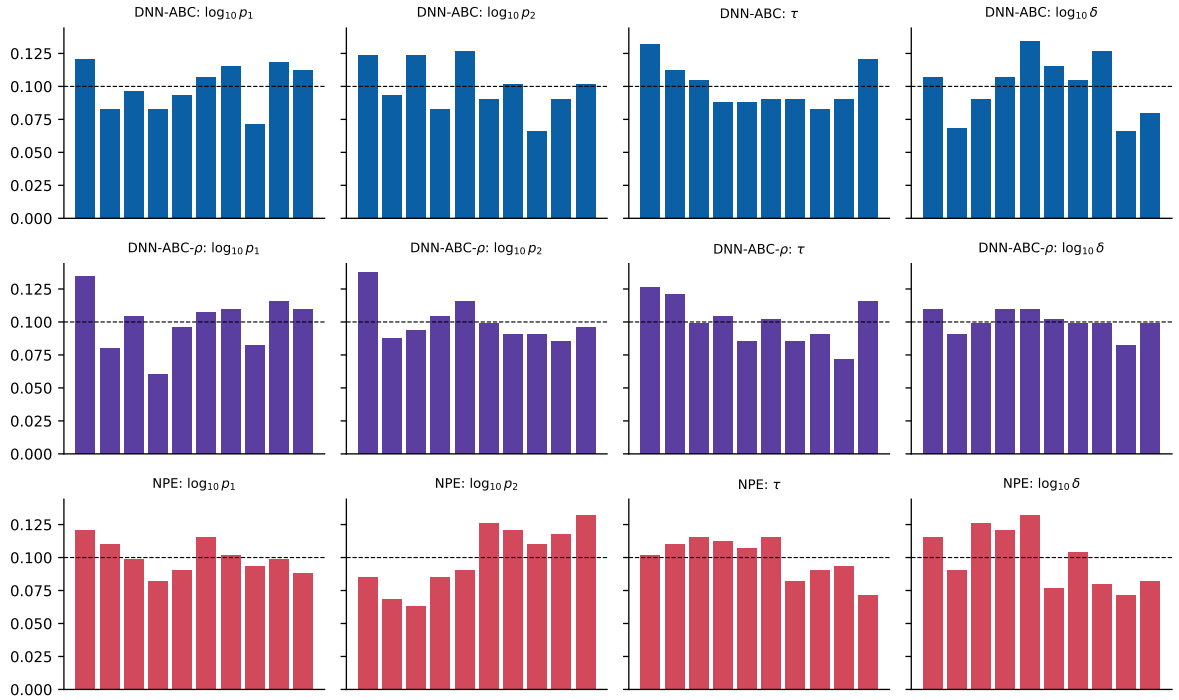


Figure S1: Simulation-based calibration rank histograms for DNN-ABC (top), DNN-ABC-ρ (middle) and NPE (bottom), on the data sets with $\bar{d} > 0$; dashed line is the uniform level.